

Task 1

```
Starting Run 1...
Run 1: hits = 20,152,358 | Pi = 1.612189 | Time = 4796.66 ms
Starting Run 2...
Run 2: hits = 19,987,383 | Pi = 1.598991 | Time = 4623.47 ms
Starting Run 3...
Run 3: hits = 19,923,420 | Pi = 1.593874 | Time = 4643.56 ms
Starting Run 4...
Run 4: hits = 20,003,450 | Pi = 1.600276 | Time = 4622.32 ms
Starting Run 5...
Run 5: hits = 19,796,825 | Pi = 1.583746 | Time = 4605.18 ms
```

Run	Pi
1	1.612189
2	1.598991
3	1.593874
4	1.600276
5	1.583746

Task 2

```
Single Thread
Pi: 3.141538
Time: 5884.75 ms

4 Threads + Lock
Pi: 3.141945
Time: 10431.26 ms
```

Version	Threads	π	Time
Single Thread	1	3.141538	5884.75 ms
Synchronized	4	3.141945	10431.26 ms

Task 3

```
Running 1 process(es)...  
1 processes | Time = 14997.39 ms | Pi = 3.141927  
Running 2 process(es)...  
2 processes | Time = 9030.38 ms | Pi = 3.141610  
Running 4 process(es)...  
4 processes | Time = 5420.85 ms | Pi = 3.141778  
Running 8 process(es)...  
8 processes | Time = 4416.27 ms | Pi = 3.141721  
Running 16 process(es)...  
16 processes | Time = 3495.06 ms | Pi = 3.141506  
Running 32 process(es)...  
32 processes | Time = 3735.52 ms | Pi = 3.141586
```

Processes	Runtime (ms)	Speedup	Efficiency
1	14997.39	1.00×	100.00%
2	9030.38	1.66×	83.04%
4	5420.85	2.77×	69.17%
8	4416.27	3.40×	42.45%
16	3495.06	4.29×	26.82%
32	3735.52	4.01×	12.55%